

Date: 27/02/25

Unit - 2

Fuel

Q1 Define Fuels and classify the coal on the basis of carbon content.

Ans Any material which produces a large amount of heat energy on complete combustion, the heat energy can be utilized for domestic & industrial purpose. This material is known as Fuel.

On the basis of carbon content, coal has four types -

(i) Peat $\Rightarrow$ 57 to 60% of carbon presence
lowest category of coal.

(ii) Lignite $\Rightarrow$ 60 to 70% of carbon presence
more compact than peat.

(iii) Bituminous $\Rightarrow$ 75 to 90% of carbon presence
most commonly used coal.

(iv) Anthracite $\Rightarrow$ 90 to 95% of carbon presence
best quality of coal.

Q2 What are good fuel's characteristics?

Ans A Good fuel contains -

- (i) low quantity of moisture, volatile matter.
- (ii) high calorific value, burns easily, without
- (iii) producing smoke or harmful gases.
- (iv) Should be sustainable & renewable.
- (v) Should be safe & pollution free.

Q3 What are gross & net calorific values?

Ans Net calorific value is the amount of heat produced after combustion, when volatile matter / gases are allowed to escape in the atmosphere (LCV)

Gross calorific value is the amount of heat produced by combustion of fuel & gases & volatile matter is allowed to cool is known as gross calorific value (HCV)

$$HCV = LCV + \text{Heat of vaporization}$$

Q4 What is pulverisation of coal? write its significance

Ans Coal is converted into powder form is known as pulverisation.

It increase surface area of coal & burning becomes easier.

Q5 Explain ultimate ^{Analysis} process of coal.

Ans It consists of determination of combustion value, ash, moisture, Volatile matter.

* Moisture Content:- Powdered coal is taken in silica crucible & heated & cooling for 1 hr.

$$\% \text{ of moisture} = \frac{\text{loss in weight}}{\text{weight of coal}} \times 100$$

(ii) Volatile content:- moisture free coal heated in muffle furnace for 7 to 8 minute.

$$\% \text{ Volatile} = \frac{\text{loss in weight} \times 100}{\text{weight of coal}}$$

(iii) Ash:- Volatile free coal heated at 700-750°C in presence of oxygen.

$$\% \text{ ash} = \frac{\text{weight of ash} \times 100}{\text{weight of coal}}$$

(iv) Fixed Carbon:- volatile free coal heated. Remaining material, after removal of volatile matter, Ash, moisture, is known as fixed carbon.

$$\% \text{ Fixed carbon} = 100 - (\% \text{ moisture} + \% \text{ volatile} + \% \text{ Ash})$$

Significance :-

It provide information about quality of coal.

→ If % of fixed carbon is high then it has high calorific value which mean it's a good quality coal.

→ If % of moisture & volatile matter is low then it's a good quality of coal.

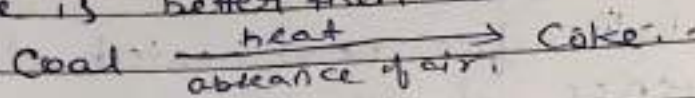
→ If Ash is less the coal is of a good quality.

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Q6 What is carbonation? Explain recovery of by product of Otto-Hoffmann's by product method.

Ans Coal is heated 'strongly' in absence of air, it converts into coke.

Coke is better than coal.



It is known as carbonation.

* Recovery of by product of Otto-Hoffmann's by product method

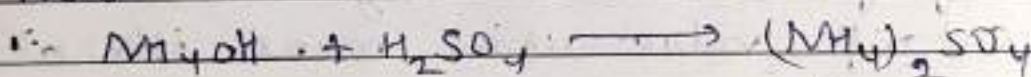
It contains tar, ammonia, H_2S , naphthalene and moisture.

1. Tar:- Coke process passed through NH_3 (Ammonia gas) and it separates tar.

(2) Ammonia:- by spreading water.



Now



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used as fertilizer.

(3) Naphthalene:- The gases are passed where water at low temperature is sprayed.

Naphthalene get condensed.

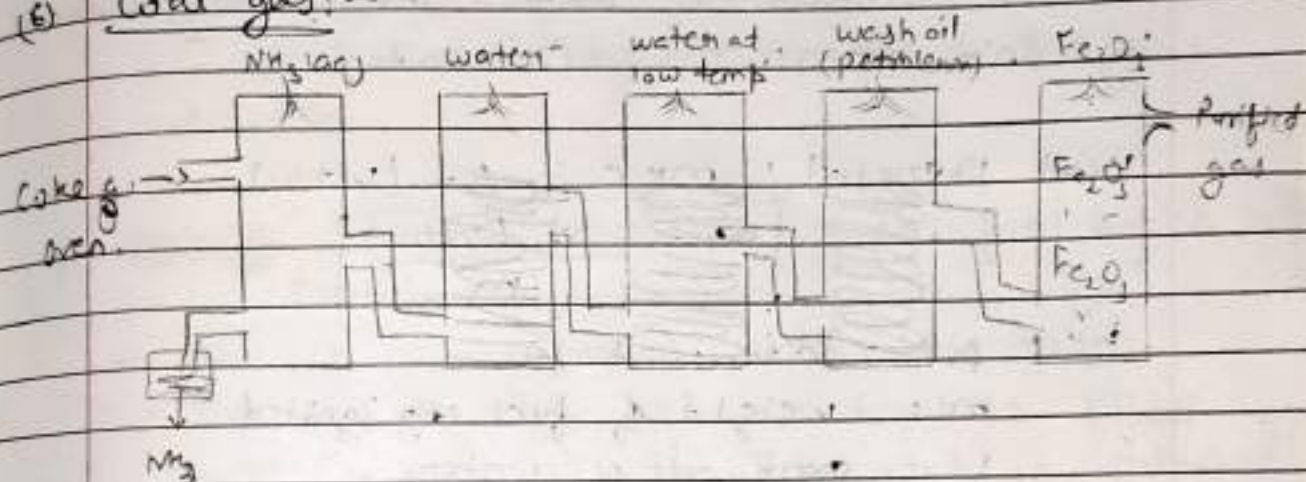
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(4) Benzene:- The sweet part is passed through petroleum.

(5) H₂S:- Passing through moist Fe₂O₃



(6) Coal gas:-



Q7 Why is coke preferred in metallurgical process?

Ans Coke is preferred in metallurgical process because

(i) It has high carbon content. It helps in reducing metal oxides to their respective values.

(ii) It has high calorific value.

(iii) It is porous structure.

(iv)

Q8 Describe Bomb calorimeter?

Ans Bomb calorimeter is used for solid and non-volatile liquids for calculating the calorific values.

Principle A known amount of fuel is burnt the heat produced is absorbed by known amount of water. The heat produced by unit mass/v is calculated.

$$\text{Heat liberated} = m \times c = \Delta t (W + w)$$

$$c = \frac{\Delta t (W + w)}{m}$$

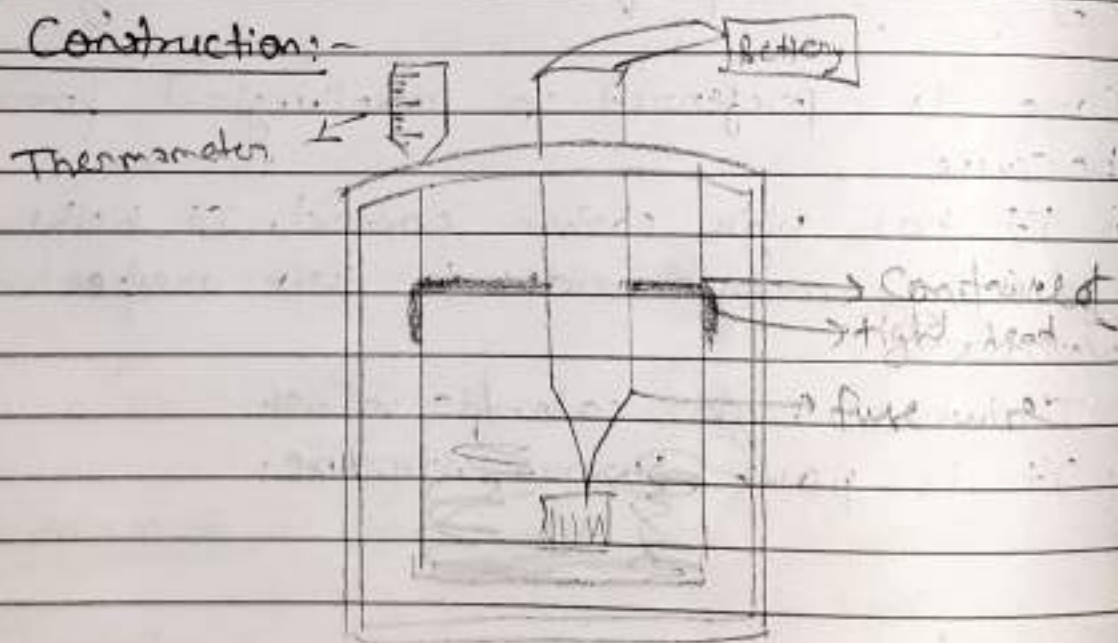
here Δt = diff in temp.

m = weight of fuel ing ignited

W = amount of water.

w = water eq. of calorimeter.

Construction:-



Structure

- i) Stained steel bomb :- It contains given sample of fuel.
- ii) Copper calorimeter :- It contains a thermometer which is an electrically operated.
- iii) Air Jacket & water Jacket :- It cover to prevent any heat losses due to radiation.

Working :- A known mass of given fuel taken in crucible.

- > Crucible is supported over the ring.
- > lid is tightly screwed bomb is filled with O_2 at 25 atm. Fuse wire is stretched across the electrodes.
- > It is lowered down in copper calorimeter (water). It is connected to 6V battery & circuit is done. Now sample starts burning & heat is liberated & temp is measured.

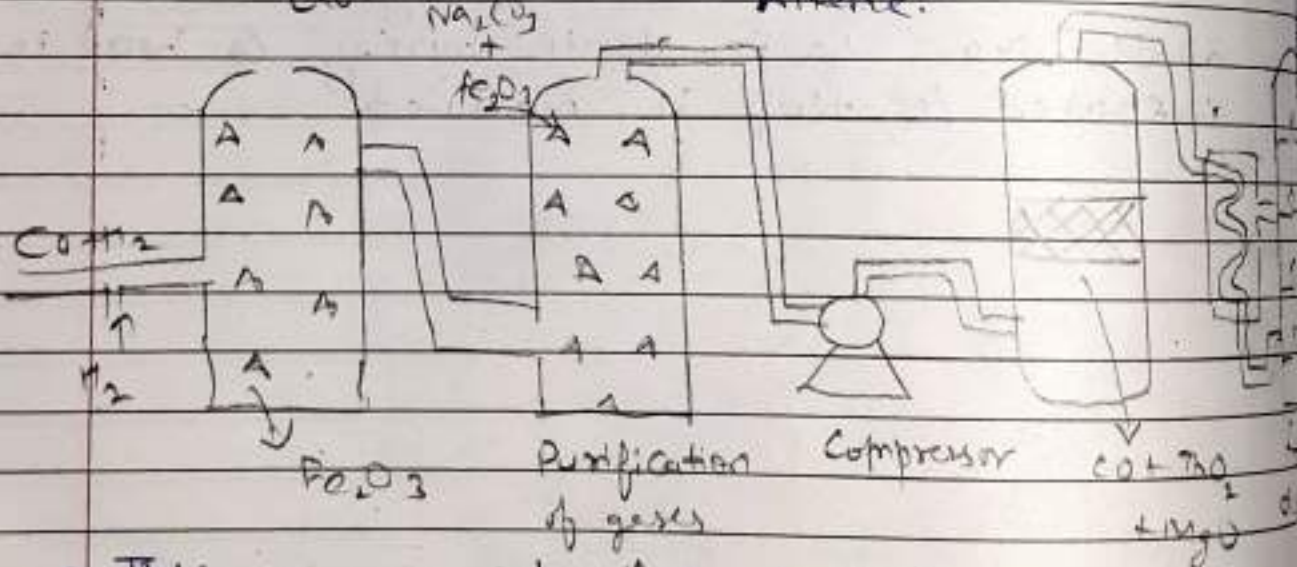
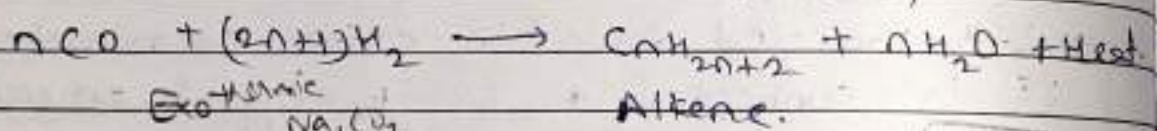
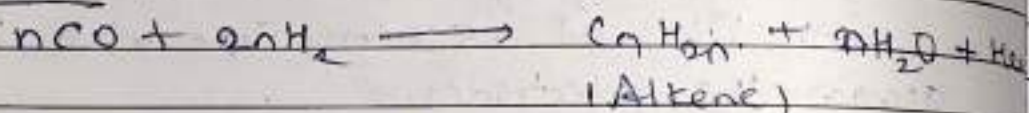
Q.9. What is synthetic petrol? Explain Fischer-Tropsch method.

Ans. When coal & its products are hydrogenated they are converted into petroleum like substance at appropriate temp & pressure. This petroleum like substance is known as synthetic petrol.

Fischer-Tropsch process:

$\text{CO} + \text{H}_2$ (water gas) passed over catalyst at high temperature and pressure.
Zinc, Cr_2O_3 , NiO these are the catalysts which are used.

Reaction



- This is an expensive process.
- Useful in emergency.

Q.10

Ans

What is reforming? Write its advantages?
 Reforming is used in petroleum & petrochemical industries to improve the quality of fuel by converting lower-grade hydrocarbons into more valuable products. It typically involves breaking & rearranging hydrocarbon molecules to increase the octane number of gasoline or produce hydrogen-rich gases.

Advantage

- (i) It increases octane rating of fuel & reduces engine knocking.
- (ii) Generates hydrogen, which is essential for refining processes like cracking, desulfurization.
- (iii) Improves yield of valuable products.
- (iv) Enhance fuel efficiency.
- (v) Reduces environmental impact.

Q.11

Ans

What is sweetening of Petrol?
 Sweetening is a refining process used to improve the quality of petrol (gasoline) by reducing its sulfur content & removing foul-smelling sulfur compound like mercaptans.



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Disulphide

Q12 What is octane number. How it can be increased in fuel?

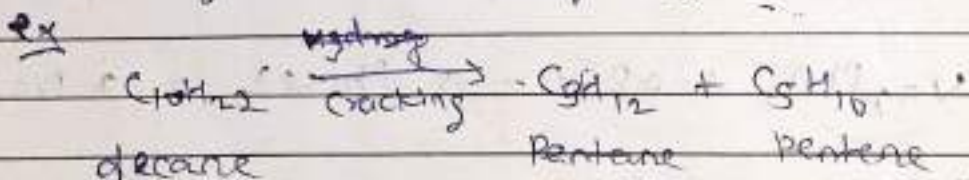
Ans Octane number is a measure of a fuel's ability to resist knocking or pre-detonation in an internal combustion engine. Higher octane number indicates better resistance of knocking & better quality of fuel.

Octane Number can be increased by several methods in fuel.

- (i) Blending with high octane components. ex benzene, toluene, ethanol etc
- (ii) Refining Process - like catalytic reforming, isomerization or alkylation.
- (iii) Using octane Boosting Additives (Anti knocking agents) like TEL, MTBE, E10, E85 etc.

Q13 What is cracking, explain Fixed bed catalytic cracking?

Ans Decomposition of Higher hydrocarbon molecules into lighter and useful hydrocarbons.



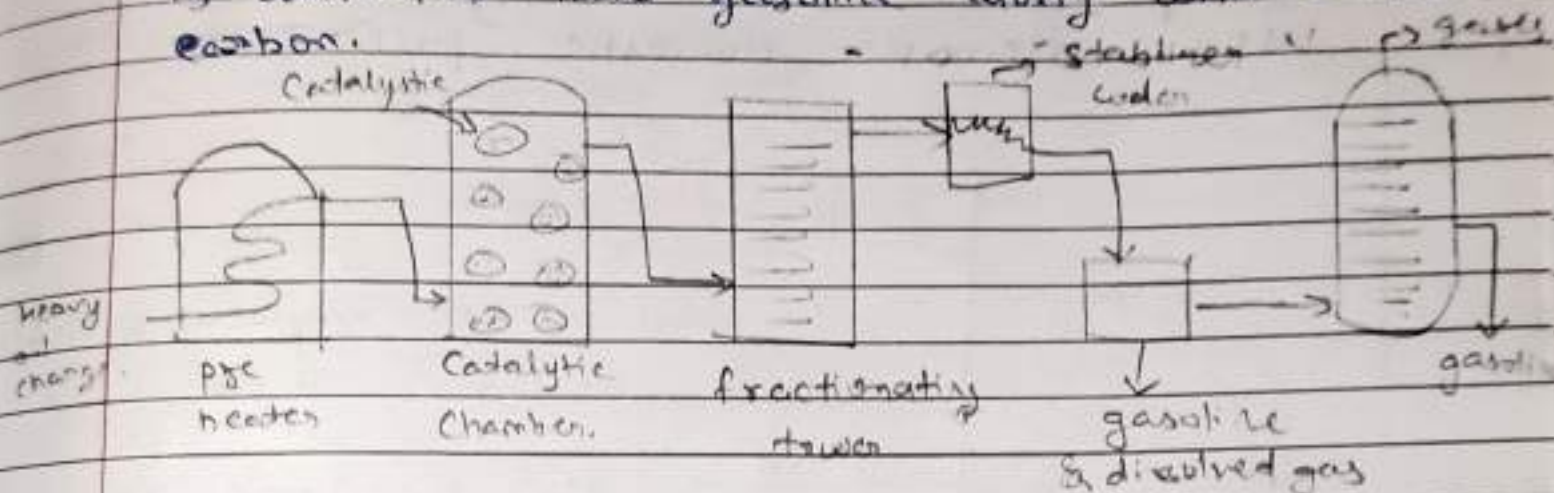
It has two types -

- (i) Thermal cracking (ii) Catalytic cracking

Fixed bed catalytic cracking:

In this method

Heavy oil vapours are preheated ~~catalytic~~ at 425° to 450°C and passed through catalytic chamber. About 40% of charged heavy oil is converted into gasoline along with 2-4% carbon.



Q14 Write a short note on fractional distillation?

Ans Crude oil is heated in a furnace at temp. of 400°C by which most of the oil is vaporised. These vapours are passed through distillation tower in which temp. decreased from top to bottom of tower.

Principle The fraction having lowest boiling point condense near the top of the tower.

Various fractions are collected according to their boiling points.

iv Gasoline $\rightarrow$ having low boiling point.

Calorific value = 11250 kcal/kg

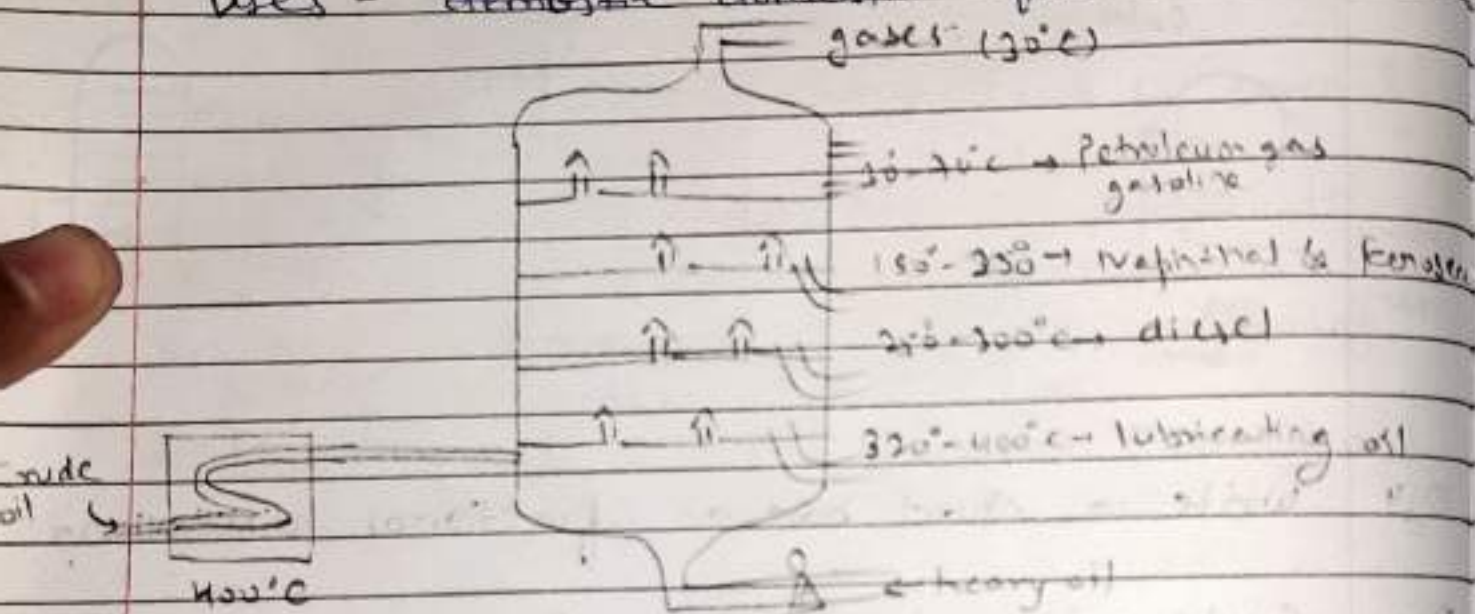
Used as fuel in engines of automobile & aeroplanes.

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Kerosene oil calorific value = 11100 kcal/kg
Used as good fuel for jet engines.

Liquidified petroleum gases - Light hydro-carbons
such as propane & butane & can be
stored in containers under pressure.
Calorific value = 27500 kcal/kg
Uses - domestic fuel.
gases (30°C)



Q15. What is producer gas? Explain Junker's calorimeter.